Why Did Operational Throughput Drop Last W

A Data-Driven Investigation into Mid-Year Flow Rate Fluctuations

Targeting Logistics, Weather, and Governance across Nairobi, Mombasa & Eldoret Terminals

THE OPERATIONAL PROBLEM

During the week of June 25 to July 1, 2026, the central operations console flagged a significant reduction in mean flow rates (LPM) at several key depots, threatening daily distribution targets.

Key Operational Questions Investigated:

- Environmental Impact: Did heavy precipitation trigger pump line friction or viscosity drops?
- Workforce Governance: Did team sizes or supervisor experience on the holiday night shift affect throughput?
- Machine Stability: Did localized pressure limits (100 - 300 PSI) remain within stable thresholds?

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The Data Integration Architecture

PIPELINE WORKFLOW: CSV + REST API + SQLITE DB

We developed a robust ETL pipeline merging three independent data sources into a standardized, date-aligned master hourly dataset.

1. Internal CSV Source

File: ops_sensor_log_cleaned.csv

- Pre-cleaned dataset
- Loaded directly in pandas
- Chronological sorting
- Hourly resampling

2. External API Source

REST Endpoint: Open-Meteo API

- Hourly Nairobi Weather
- Rain, Temp, Humidity
- Connection timeout handling
- Offline synthetic fallback

3. Relational Database

Engine: SQLite via SQLAlchemy

- holiday_calendar (Table)
- shift_details (Table)
- SQL JOIN and GROUP BY query
- Loaded via pd.read_sql

Unified Date-Aligned Master DataFrame

- Operational logs resampled to hourly means by Zone & Shift
- Merged with Weather on timestamp (Many-to-One)
- Merged with SQLite on Date, Zone, and Shift (Many-to-One)
- Missing values resolved using linear ffill() and bfill()

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Insights & Strategic Recommendations

KEY FINDINGS FROM MULTI-SOURCE ANALYSIS

1. The Weather Myth

Statistical Correlation r = 0.00

Our Pearson correlation matrix shows that rainfall has exactly r = 0.00 correlation with mean flow rates.

Insight:
Depot pumps and lines are physically shielded, meaning precipitation does NOT directly retard pipe hydraulics.

2. Governance & Staffing

Night Shift Capacity Cut by 50%

Night shifts operate with a planned team size of 4, compared to 8 on Morning shifts.

Insight:
On holidays and rainy nights, this capacity drop leads to delayed maintenance recovery, resulting in prolonged system throttling.

3. Actionable Strategy

Director Recommendations

Operational changes to make:

1. Weather-Adaptive Staffing:
Increase holiday night shifts from 4 to 6 when storm warnings are active.

2. Supervisor Reallocation:
Assign top performing leads (Munyao, Njoroge) to holiday night shifts during wet seasons.

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